

RENJU THOMSON

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EDUCATION

University of Washington, Seattle

Sep 2024 – Aug 2026

Master of Science in Electrical and Computer Engineering | GPA: 3.8

Coursework: Database Management Systems, Machine Learning, Data Structures & Algorithms, Object-oriented Design

APJ Abdul Kalam Technological University, India

Aug 2018 – Aug 2022

Bachelor of Technology in Electronics and Communication Engineering | GPA: 3.7

- Vice President, IEEE Branch - Organized tech workshops and alumni events that introduced students to AI/ML fundamentals and expanded early exposure to diverse career paths

EXPERIENCE

Software Engineer Co-op | Nokia, Sunnyvale, CA

Jan 2026 – Present

- Designed and automated an end-to-end **ETL pipeline** that extracted, cleaned and transformed raw coverage metrics across 7+ software modules into structured JSON datasets, cutting manual reporting effort by **40%**.
- Built an interactive **analytics dashboard** with KPI cards, historical trend charts and module-level drill-downs, giving 30+ engineers and stakeholders self-serve visibility into weekly coverage metrics.
- Redesigned the **PostgreSQL** database for a multi-agent testbed assistant normalizing 10+ tables, rebuilding lookup logic and migrations scripts, enabling **700+** engineers to retrieve hardware information in seconds with **50%** improved reliability.
- Replaced manual, fragmented data collection by building an automated **data ingestion** pipeline that ingests **5000+** testbed records from GitLab and REST APIs into a PostgreSQL with SHA-256 change detection, delivering accurate query-ready data.
- Developed **time-series analysis** with automated **anomaly detection** for weekly software test-coverage metrics, flagging threshold-based regressions and triggering email alerts, catching coverage drops within minutes instead of during weekly manual reviews and saving developers **5 hours/week**.
- Collaborated in **cross-functional teams** to gather requirements to develop **2+ user-centric AI agent skills** using **prompt engineering** tailored to team-specific workflows, incorporating iterative feedback to improve adoption and usability.

Data Analyst | NASA Jet Propulsion Laboratory - UW Capstone Research, Seattle

Jan 2025 – Jun 2025

- Collected, cleaned and analyzed **25+** years of large-scale NASA multi-hazard datasets to build scalable and efficient data solutions for modeling the impact of extreme weather events on U.S power grids.
- Led exploratory data analysis (**EDA**) using Python to identify patterns and trends and effectively communicated insights through interactive **visualizations** (Tableau, Plotly, Matplotlib) to non-technical stakeholders.
- Identified highest joint risk areas in U.S by performing advanced statistical modeling (multivariate analysis, EVA) on wildfire, drought and wind datasets through **geospatial visualizations** for better risk management.

Analyst Trainee | Caritas Institute, India

Mar 2023 – Jun 2024

- Built **dashboards** and **reports** to visualize key hospital metrics (patient load, discharge rates, bed occupancy), enabling administrators to monitor performance trends and make timely, data-driven decisions across five departments.
- Identified overstocking of low-demand drugs by analyzing monthly medication usage trends using **SQL and Power BI** and recommended revised ordering thresholds, achieving a **15%** reduction in inventory costs.

ACADEMIC PROJECTS

Speech Emotion Recognition | Machine Learning

May 2025 – Jun 2025

- Developed a lightweight Convolutional Neural Network (CNN) model on embedded hardware, achieving **92.7%** accuracy for real-time emotion recognition across five emotion classes (Happy, Sad, Angry, Fear, Silence).
- Presented findings and recommendations in team meetings, applied feedback from mentors to improve data preparation approach and adjust model parameters for **higher accuracy**.

Superhero Classification

Nov 2024 – Dec 2024

- Led a team of five to build and train a CNN model to classify superheroes, achieving **100%** validation accuracy using transfer learning while reducing misidentification.
- Troubleshoot** model errors by analyzing outputs to find improvement opportunities and showed initiative to learn by exploring new classes and refining costume-based inputs for better generalization.

TECHNICAL SKILLS

- Programming Languages:** Python, SQL, C++, Bash/Shell Scripting
- Data Analysis:** ETL, Data Modeling, Data Cleaning/Wrangling, Data Mining, A/B Testing, Predictive Modeling
- Python Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Plotly
- Data Visualization:** Tableau, Power BI, Microsoft Excel(PivotTables, VLOOKUP)
- Databases:** PostgreSQL, MySQL, SQLite, NoSQL, pgvector, Docker-based DB management
- Tools & Platforms:** Jupyter, Git, GitLab, GitHub, Docker, Linux, Azure, VS Code, PgAdmin4, Cursor, GitHub Copilot
- AI & LLMs:** Prompt Engineering, LangChain, LanGraph (ReAct Agents), LLM Integration, Technical Documentation